

## OVERVIEW

The MOT0110\_0 PhidgetSpatial Precision 3/3/3 is a nine-degree-of-freedom inertial and magnetic sensing board. It combines a three-axis accelerometer, three-axis gyroscope, and three-axis magnetometer. Applications can read each sensor independently or use the synchronized Spatial channel and the built-in AHRS or IMU algorithms to obtain orientation as quaternions.

**Official source:** <https://www.phidgets.com/?prodid=1205>

This reference condenses the official product description, user-guide notes, specifications, data-interval rules, software channels, and device history. Consult the current Phidget22 documentation before building safety-critical navigation or control systems.

## At a glance

| Item                             | Value                                         |
|----------------------------------|-----------------------------------------------|
| <b>Product</b>                   | PhidgetSpatial Precision 3/3/3                |
| <b>SKU</b>                       | MOT0110_0                                     |
| <b>Sensors</b>                   | 3-axis accelerometer, gyroscope, magnetometer |
| <b>Acceleration range</b>        | +/-16 g                                       |
| <b>Angular-rate range</b>        | +/-2000 degrees/s                             |
| <b>Magnetic-field range</b>      | +/-8 G                                        |
| <b>Orientation output</b>        | Quaternion via AHRS or IMU algorithm          |
| <b>Connection</b>                | Direct USB or VINT Hub                        |
| <b>Driver</b>                    | Phidget22                                     |
| <b>Fastest nominal sampling</b>  | 1 ms/sample, connection dependent             |
| <b>Temperature stabilization</b> | Sensor heating to 50 degrees C                |

## What the sensors measure

|               |                                                                                                                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Accelerometer | Reports acceleration on X, Y, and Z. A stationary sensor also measures gravity, so tilt can be estimated when other acceleration is small. During motion, the output includes gravity, linear acceleration, vibration, and noise. |
| Gyroscope     | Reports angular velocity around X, Y, and Z. Integrating angular velocity provides short-term orientation changes, but bias and noise cause drift over time.                                                                      |
| Magnetometer  | Reports magnetic-field strength on X, Y, and Z. After calibration it can provide a heading reference, but nearby steel, magnets, motors, high-current wiring, and electrical equipment can distort the field.                     |
| Spatial       | Reports synchronized accelerometer, gyroscope, and magnetometer data with a common timestamp. It also supports AHRS and IMU orientation algorithms and quaternion output.                                                         |
| Temperature   | Reports internal temperature. The board can heat the sensors to 50 degrees C to reduce temperature-related bias changes.                                                                                                          |

## AHRS and IMU modes

|      |                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AHRS | Combines accelerometer, gyroscope, and magnetometer data. The magnetic reference can correct yaw drift, but magnetic disturbance or poor compass calibration can degrade heading.    |
| IMU  | Uses inertial measurements without relying on the magnetometer for heading correction. This is useful in magnetically disturbed environments, but absolute yaw will drift with time. |

Choose the mode for the environment and application. Quaternions avoid the singularity associated with Euler angles, although pitch/roll/yaw can still be derived for display. Treat the initial orientation, coordinate conventions, and quaternion component order as part of the software interface; verify them with the current Spatial API documentation.

## Connection and setup

|            |                                                                                                                                                                                                                                                                                                                              |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct USB | Connect the board to the host with a supported USB cable. USB operation is full-speed, with a specified supply range of 4.1-5.3 V DC.                                                                                                                                                                                        |
| VINT       | Connect the board to a VINT Hub using a three-wire Phidget cable. Maximum supported VINT communication speed is 1 Mbit/s, but the selected hub and configuration determine the actual speed. The official page notes that HUB0000 and HUB5000 operate at 100 kbit/s, while HUB0001 can be configured as high as 1000 kbit/s. |
| Software   | Install the current Phidget22 library, open the Phidget Control Panel, and confirm that the accelerometer, gyroscope, magnetometer, Spatial, and temperature channels appear.                                                                                                                                                |

## Commissioning checklist

1. Mount the board rigidly, document the axis directions, and keep it away from vibration sources when possible.
2. Choose USB or VINT and confirm the connection can support the intended channels and data intervals.
3. Start the Phidget Control Panel and check all three sensor axes while rotating the board by hand.
4. Enable temperature stabilization when repeatable precision is important. Allow the internal temperature to reach and settle near 50 degrees C before bias-sensitive measurements.
5. Calibrate the magnetometer in its final installation, away from temporary magnetic objects. Recalibrate after changing the enclosure, fasteners, wiring, battery, motor, or mounting location.
6. Select AHRS or IMU fusion, establish the initial orientation, and verify quaternion and axis conventions with known rotations.
7. Select a data interval appropriate to the motion bandwidth, connection, number of open channels, processor load, and noise requirements.
8. Log timestamps and status/error events. Test recovery from disconnection, out-of-range data, magnetic disturbance, and startup transients.

## Specifications

### Accelerometer

| Property                                | Specification |
|-----------------------------------------|---------------|
| <b>Measurement range</b>                | +/-16 g       |
| <b>Resolution</b>                       | 30 ug         |
| <b>Bandwidth</b>                        | 250 Hz        |
| <b>Noise at 1 ms</b>                    | +/-3 mg       |
| <b>Noise at 10 ms</b>                   | +/-1.5 mg     |
| <b>Noise at 100 ms</b>                  | +/-200 ug     |
| <b>Maximum drift</b>                    | 2 mg          |
| <b>Bias, +15 to +30 degrees C</b>       | 5 mg          |
| <b>Bias, stabilized at 50 degrees C</b> | 1 mg          |

**Gyroscope**

| Property                         | Specification      |
|----------------------------------|--------------------|
| Angular-rate range               | +/-2000 degrees/s  |
| Resolution                       | 0.004 degrees/s    |
| Noise at 1 ms                    | +/-0.2 degrees/s   |
| Noise at 10 ms                   | +/-0.05 degrees/s  |
| Noise at 100 ms                  | +/-0.015 degrees/s |
| Maximum drift                    | 0.1 degrees/s      |
| Bias, +15 to +30 degrees C       | 0.5 degrees/s      |
| Bias, stabilized at 50 degrees C | 0.05 degrees/s     |

**Magnetometer**

| Property             | Specification |
|----------------------|---------------|
| Magnetic-field range | +/-8 G        |
| Resolution           | 1.5 mG        |
| Noise at 10 ms       | +/-10 mG      |
| Noise at 50 ms       | +/-5 mG       |
| Noise at 100 ms      | +/-2.5 mG     |

**Board and environmental**

| Property                         | Specification          |
|----------------------------------|------------------------|
| Controlled by                    | VINT or USB            |
| USB stack / speed                | PHIDUSB / Full Speed   |
| Driver support                   | Phidget22              |
| Maximum VINT speed               | 1 Mbit/s               |
| Maximum current consumption      | 60 mA                  |
| Maximum current, heating enabled | 450 mA                 |
| Sampling interval range          | 1 ms to 1 s per sample |
| USB voltage range                | 4.1-5.3 V DC           |
| Operating temperature            | -40 to 85 degrees C    |
| Country of origin                | China                  |

The 1 ms minimum is conditional. The achievable interval depends on USB versus VINT, VINT speed, and how many channels are open at the same time.

**Minimum data intervals**

For direct USB, each channel uses the True Minimum row below regardless of how many channels are open. For VINT, add the table contribution for every open channel at the selected VINT speed, then add 1 ms for every open channel beyond the first. A channel cannot run faster than its True Minimum.

| VINT speed   | Accel. | Gyro | Mag. | Spatial | Temp. |
|--------------|--------|------|------|---------|-------|
| 100 kbit/s   | 3      | 3    | 3    | 8       | 2     |
| 160 kbit/s   | 2      | 2    | 2    | 6       | 1     |
| 250 kbit/s   | 1      | 1    | 1    | 4       | 1     |
| 500 kbit/s   | 1      | 1    | 1    | 3       | 1     |
| 1000 kbit/s  | 1      | 1    | 1    | 2       | 1     |
| True minimum | 1      | 1    | 10   | 1       | 10    |

All values are milliseconds. Example from Phidgets: with accelerometer, magnetometer, and temperature open at 160 kbit/s, the calculated interval is  $2 + 2 + 1 + 2 = 7$  ms. The accelerometer may use 7 ms, while the magnetometer and temperature channel remain limited to their 10 ms true minimum.

### Phidget22 software channels

| Channel                     | API class         | Index | Interfaces |
|-----------------------------|-------------------|-------|------------|
| <b>3-Axis Accelerometer</b> | Accelerometer     | 0     | USB, VINT  |
| <b>3-Axis Gyroscope</b>     | Gyroscope         | 0     | USB, VINT  |
| <b>3-Axis Magnetometer</b>  | Magnetometer      | 0     | USB, VINT  |
| <b>Spatial</b>              | Spatial           | 0     | USB, VINT  |
| <b>Temperature Sensor</b>   | TemperatureSensor | 0     | USB, VINT  |

The same logical channels are available through either interface. The code-sample generator on the official product page provides current examples by language and operating system. Phidgets lists examples for C, C#, Objective-C, Swift, Visual Basic, and Max, including compass calibration and Spatial AHRS/IMU examples where applicable.

### Practical measurement guidance

|              |                                                                                                                                                                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Averaging    | Longer data intervals reduce the published noise figures but reduce time resolution. Select an interval from the dynamics of the system rather than choosing the fastest rate automatically.                                                                      |
| Warm-up      | Heating improves specified accelerometer and gyroscope bias, but increases maximum current consumption from 60 mA to 450 mA. Confirm the host or hub power budget and wait for thermal stabilization.                                                             |
| Mounting     | Flexing, loose fasteners, cable forces, and vibration can contaminate readings. Rigid mounting and mechanical isolation appropriate to the target bandwidth improve repeatability.                                                                                |
| Magnetics    | Calibrate the compass after final assembly. Keep the sensor away from motors, speakers, ferrous brackets, magnets, switching converters, and high-current conductors. If the magnetic environment cannot be controlled, consider IMU fusion and accept yaw drift. |
| Aliasing     | The accelerometer bandwidth is 250 Hz, but the usable measured bandwidth is also constrained by sampling rate and filtering. Ensure the sample rate and filters are appropriate for the highest signal frequency of interest.                                     |
| Out of range | Software should handle out-of-range errors and saturated values without stopping the acquisition loop. A saturated axis does not represent the true motion beyond the sensor range.                                                                               |

### Limitations and safety

- This is a measurement component, not an independently certified safety, flight, medical, or life-support navigation system.
- Orientation accuracy depends on calibration, mechanical mounting, motion, vibration, temperature, magnetic environment, algorithm settings, and startup conditions; sensor resolution alone is not system accuracy.
- Acceleration includes gravity and other motion. A single instantaneous accelerometer sample cannot generally separate tilt from linear acceleration.
- Magnetometer heading can change near ferrous material or electrical current. Validate the installed system throughout its full operating environment.
- Provide application-level plausibility checks, logging, fault handling, and redundant sensing when incorrect orientation or motion data could cause hazardous behavior.

**Device history published by Phidgets**

| Date     | Version | Change                                                  |
|----------|---------|---------------------------------------------------------|
| Aug 2022 | 100     | Product release                                         |
| May 2023 | 104     | Auto-setSpeed; improved VINT stability and ESD immunity |
| Jan 2024 | 110     | Fixed AHRS parameter update and dataInterval issues     |

**Further reading**

- Product description, user guide, specifications, API, downloads, and current code-sample generator:  
<https://www.phidgets.com/?prodid=1205>
- From the product page, follow the Spatial, Accelerometer, Gyroscope, and Magnetometer guides, plus the VINT communication-speed and cable-length guides.

*Document status:* Prepared 11 September 2026 from the official product page. Product availability, pricing, firmware, software, and specifications can change; the official page is authoritative.